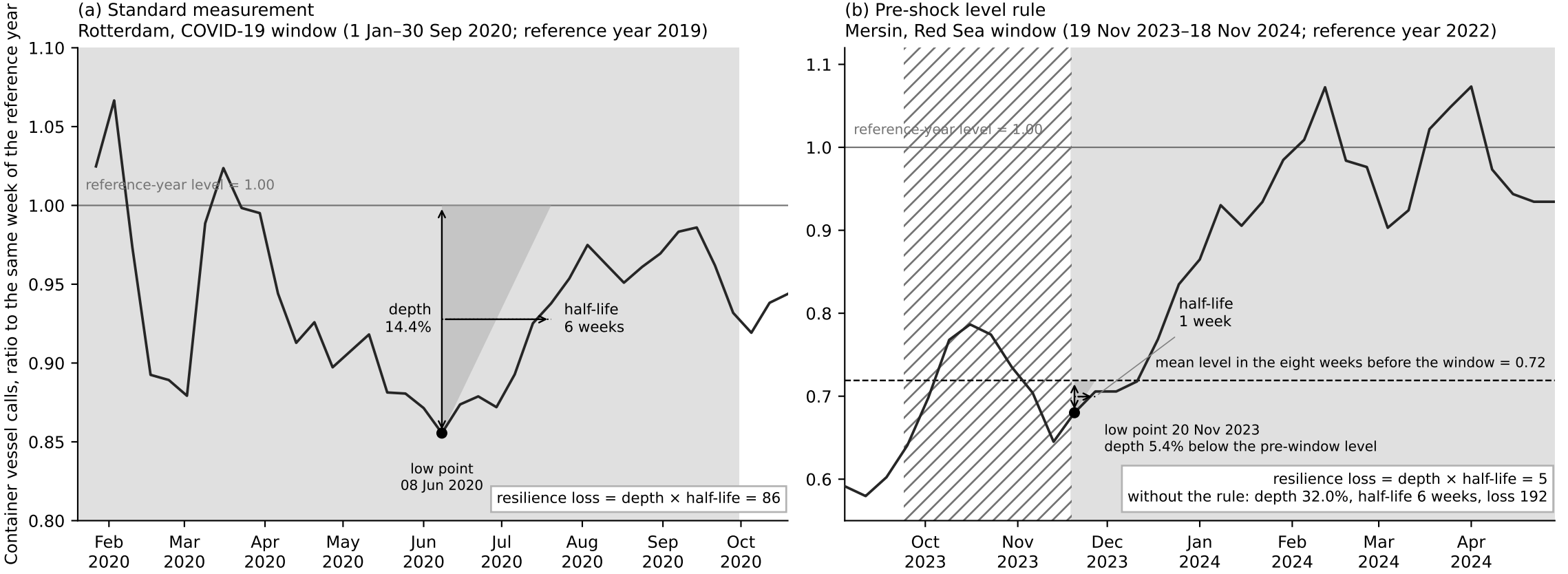
(a) Standard measurement
Rotterdam, COVID-19 window (1 Jan–30 Sep 2020; reference year 2019)
Container vessel calls, ratio to the same week of the reference year
1.10
1.05
1.00
0.95
0.90
0.85
0.80
reference-year level = 1.00
depth
14.4%
half-life
6 weeks
low point
08 Jun 2020
resilience loss = depth × half-life = 86
Feb 2020
Mar 2020
Apr 2020
May 2020
Jun 2020
Jul 2020
Aug 2020
Sep 2020
Oct 2020
(b) Pre-shock level rule
Mersin, Red Sea window (19 Nov 2023–18 Nov 2024; reference year 2022)
1.1
1.0
0.9
0.8
0.7
0.6
reference-year level = 1.00
half-life
1 week
mean level in the eight weeks before the window = 0.72
low point 20 Nov 2023
depth 5.4% below the pre-window level
resilience loss = depth × half-life = 5
without the rule: depth 32.0%, half-life 6 weeks, loss 192
Oct 2023
Nov 2023
Dec 2023
Jan 2024
Feb 2024
Mar 2024
Apr 2024